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Assessing the Limits of Algorithmic Fairness: An Empirical Evaluation of Reweighting versus Adversarial Debiasing in Credit Scoring

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Master of Science in **Advanced Computer Science**, at the University of Hertfordshire (UH).

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Glossary

Term	Definition
Adversarial Debiasing	An advanced in-processing bias mitigation architecture that utilizes competing neural networks to mathematically penalize a model for learning protected attributes.
AIF360	AI Fairness 360 (An open-source toolkit developed by IBM utilized for detecting and mitigating algorithmic bias).
DI (Disparate Impact)	A mathematical ratio used by regulatory bodies to quantify demographic fairness and measure statistical adverse impact.
EEOC	Equal Employment Opportunity Commission (The United States regulatory agency that established the 0.80 four-fifths compliance threshold).
Epoch	One complete mathematical pass of the entire training dataset through a machine learning algorithm during the optimization phase.
GAN (Generative Adversarial Network)	A class of machine learning frameworks utilizing a zero-sum game between two competing neural networks
GDPR	General Data Protection Regulation (The European Union data privacy and security framework).
Pareto Frontier	A mathematical boundary representing the absolute optimal trade-off limit between predictive accuracy and algorithmic fairness.
Proxy Variable	A seemingly neutral data feature that complex algorithms use to secretly reconstruct and discriminate against protected demographics.
Reweighting	A data pre-processing bias mitigation algorithm that dynamically assigns mathematical weights to training instances based on expected marginal probabilities.
SHAP (SHapley Additive exPlanations)	A game-theoretic explainability framework used to calculate and visualize the exact marginal contribution of individual predictive features.
SMOTE	Synthetic Minority Over-sampling Technique (A synthetic data augmentation method).
XAI (Explainable AI)	Computational methods and techniques that allow human auditors to mathematically verify and trust the internal logic of machine learning algorithms.
XGBoost (Extreme Gradient Boosting)	An highly optimized, distributed decision-tree machine learning library utilized as the unmitigated baseline classifier.

Abstract

The integration of artificial intelligence in credit scoring has introduced severe legal and commercial risks through the algorithmic amplification of historical biases. Under regulatory frameworks like the European Union AI Act, deploying prejudiced models exposes institutions to crippling penalties. This project investigates the mathematical accuracy-fairness trade-off by critically comparing two distinct bias mitigation methodologies: Reweighing (data pre-processing) and Adversarial Debiasing (model in-processing). Using the highly imbalanced UCI Adult Income dataset, an unmitigated baseline XGBoost classifier demonstrated severe gender discrimination. It recorded a Disparate Impact ratio of 0.1586, failing the 0.80 compliance threshold established by the US Equal Employment Opportunity Commission.

To address this discrimination, Method A utilized the AIF360 toolkit to dynamically adjust instance training weights. Alternatively, Method B implemented a complex architecture, deploying competing neural networks designed to mathematically penalize the prediction of protected attributes. Empirical testing demonstrated that Reweighing was the superior intervention, improving the Disparate Impact ratio to 0.4070 while preserving 84.62% predictive accuracy. Furthermore, game-theoretic SHAP explainability models provided visual proof that Reweighing successfully degraded the algorithm's reliance on gender, shifting primary feature importance toward merit-based metrics. Conversely, despite extensive hyperparameter tuning, the Adversarial Debiasing networks suffered from severe training instability, yielding inferior fairness (0.3512) and degraded accuracy (82.80%).

While neither algorithm achieved the strict 0.80 legal threshold due to the extreme inherent data imbalance, this research provides critical commercial insights. It establishes that mathematically simple pre-processing offers a significantly safer and computationally efficient solution for financial institutions. Ultimately, the findings expose the practical unreliability of complex in-processing networks and map the mathematical limits of algorithmic correction when models are trained on fundamentally unrepresentative legacy data.

1. Introduction

1.1 Problem Overview and the Mechanics of Algorithmic Bias

The financial sector has undergone a radical transformation over the past decade, shifting away from human-led underwriting toward automated, machine learning-driven credit risk assessment (Khandani, Kim and Lo, 2010). This transition offers undeniable advantages in processing speed and operational efficiency, allowing institutions to evaluate millions of loan applications simultaneously. However, delegating high-stakes financial decisions to predictive algorithms has exposed severe ethical, social, and professional liabilities.

Machine learning classifiers operate by identifying mathematical patterns within massive training datasets and projecting those patterns onto future predictions. The fundamental engineering flaw in this architecture is the assumption that historical data represents an objective ground truth. In reality, legacy financial data is heavily saturated with decades of societal prejudices, systemic inequalities, and historically biased lending practices. Consequently, when an algorithm optimizes purely for predictive accuracy using this flawed data, it inadvertently learns, codifies, and scales discrimination. This mathematical phenomenon ensures that protected demographic groups particularly women and ethnic minorities face systematically unfair rejection rates, regardless of their individual financial merits or creditworthiness (Barocas and Selbst, 2016; O'Neil, 2016).

1.2 Economic Impact and Commercial Risk

Beyond the immediate social harm of restricting economic mobility for marginalized groups, algorithmic bias introduces acute commercial risks to the deploying institutions. From a purely financial perspective, a model that systematically denies capital to capable, unprivileged individuals actively damages the lender. It artificially restricts the institution's viable market share and misallocates available credit resources, resulting in a measurable degradation of long-term profitability (Kozodoi, Jacob and Lessmann, 2022). Furthermore, as consumer awareness regarding algorithmic fairness grows, institutions deploying biased systems risk catastrophic reputational damage and the subsequent loss of consumer trust in a highly competitive market. Real-world failures, such as Amazon famously scrapping its proprietary AI recruiting tool after it was caught systematically penalizing female applicants, perfectly illustrate the catastrophic reputational damage of deploying unmitigated algorithms (Dastin, 2018).

1.3 Legal Frameworks and Professional Compliance

To combat these commercial and ethical risks, international regulatory bodies have introduced aggressive legal frameworks, transforming algorithmic fairness from a theoretical

academic ideal into a strict compliance mandate. Under the recently enacted European Union Artificial Intelligence Act, credit scoring and consumer risk assessment algorithms are explicitly classified as 'High-Risk AI' systems (European Parliament and Council, 2024). Financial institutions operating these systems are now legally required to implement continuous algorithmic auditing and active bias mitigation protocols. This regulatory pressure is compounded by the General Data Protection Regulation (GDPR), which establishes strict limitations on automated decision-making and demands structural accountability for how consumer data is processed (European Parliament and Council, 2016; Goodman and Flaxman, 2017). Failure to comply exposes organizations to paralyzing financial penalties and severe legal action.

To translate these broad legal requirements into operational mathematical thresholds, financial technology auditors rely heavily on the 'Disparate Impact' metric. This standard originates from the United States Equal Employment Opportunity Commission's 'four-fifths rule' (EEOC, 1978), which dictates that the selection rate for any protected demographic operating below 80% of the highest selected group constitutes statistical evidence of adverse impact. Therefore, developing computational interventions capable of satisfying this strict 0.80 threshold without destroying the algorithm's baseline predictive accuracy has emerged as a critical engineering challenge for modern computer science.

1.4 Aims, Objectives, and Research Question

The primary aim of this investigative project is to address the critical intersection of predictive computer science, commercial risk, and algorithmic ethics outlined above. To achieve this, the core objective is to critically compare the mathematical efficacy of two distinct bias mitigation architectures. The first is 'Reweighing', a foundational data pre-processing technique designed to sanitize training data before model ingestion (Kamiran and Calders, 2012). The second is 'Adversarial Debiasing', an advanced in-processing approach that utilizes competing neural networks to actively penalize a model for learning protected attributes (Zhang, Lemoine and Mitchell, 2018).

This comparative analysis seeks to answer a specific, high-value research question: To what extent can the computationally heavy Adversarial Debiasing architecture outperform the traditional Reweighing method in reducing Disparate Impact against unprivileged groups, without causing a severe degradation in the model's overall predictive accuracy?

1.5 Scope and Feasibility

The scope of this investigation is defined by the utilization of the UCI Adult Income Dataset. While serving as a widely recognized proxy for financial credit scoring, this dataset was selected because it presents an extreme, mathematically realistic class imbalance regarding

gender and income. This provides the necessary data volume to train dual neural networks without relying on artificial upsampling techniques that could skew the results.

The feasibility of this research is grounded in the use of the AI Fairness 360 (AIF360) toolkit (Bellamy et al., 2018), an industry-standard library developed by IBM that standardizes the calculation of the Disparate Impact ratio. By enforcing strict boundaries around the two specific algorithms tested and utilizing standardized evaluation metrics, the project guarantees a highly controlled, scientifically repeatable comparative environment.

1.6 Project Novelty and Innovation

Standard classification projects traditionally conclude the moment a model achieves high predictive accuracy. The novelty of this investigation lies in its explicit focus on the 'Accuracy-Fairness Trade-off' and the internal mechanics of algorithmic unlearning. Rather than treating the symptoms of bias post-prediction, this project targets the fundamental internal state of the algorithms.

Furthermore, this research innovates by integrating Explainable AI (XAI), specifically utilizing SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017). By mapping game-theoretic values to individual feature importance, this project moves beyond standard black-box fairness metrics. SHAP provides concrete visual evidence of how different architectures alter their internal logic to satisfy legal constraints, allowing for a precise evaluation of whether a model has genuinely unlearned its reliance on a protected attribute like gender, or simply hidden it behind proxy variables.

1.7 Report Structure

To systematically address these objectives, the remainder of this report is structured as follows:

- Chapter 3 (Literature Review): Provides a critical synthesis of existing academic research on algorithmic fairness, evaluating mitigation strategies and identifying specific gaps regarding the practical stability of adversarial networks.
- Chapter 4 (Methodology): Details the project design, defining the data pre-processing pipelines, the baseline testing parameters, and the specific mathematical architectures of the comparative machine learning models.
- Chapter 5 (Quality and Results): Presents the empirical findings, utilizing visual SHAP explainability and performance metrics to critically analyze the observed accuracy-fairness trade-offs.

- Chapter 6 (Evaluation and Conclusion): Delivers a comprehensive evaluation of the project's success, reflecting on project management execution, commercial implications, and outlines data-driven recommendations for future research in responsible artificial intelligence.

2. Literature Review

2.1 The Evolution of Credit Risk Assessment and Algorithmic Bias

The modernization of the financial sector has been largely defined by the aggressive adoption of machine learning architectures to automate consumer credit risk assessment. Early benchmarking studies, such as those conducted by Khandani, Kim and Lo (2010) and later updated by Lessmann et al. (2015), demonstrated that complex, non-linear classifiers particularly ensemble methods and neural networks vastly outperform traditional logistic regression in predicting loan defaults. Consequently, human-led underwriting has been systematically replaced by automated, data-driven pipelines designed to optimize purely for predictive accuracy and institutional profit.

However, delegating high-stakes socioeconomic decisions to predictive algorithms assumes that the historical data used to train these models represents an objective, unbiased ground truth. Recent computer science literature has aggressively dismantled this assumption. O'Neil (2016) characterizes these models as opaque mathematical engines that inherently scale the prejudices of their creators. Within the context of lending, the foundational training datasets are saturated with decades of societal inequalities and discriminatory financial practices.

Barocas and Selbst (2016) provided the foundational technical framework for understanding how machine learning models weaponize this historical data. They demonstrated that algorithms do not require explicit demographic inputs, such as gender or race, to execute discriminatory logic. Because machine learning classifiers are designed to find latent correlations, they efficiently reconstruct protected attributes through highly complex proxy variables. For example, an algorithm might learn to penalize specific employment sectors or geographic zip codes that heavily correlate with minority populations. Mehrabi et al. (2021) categorize this specific phenomenon as 'historical bias', noting that when a classifier accurately maps a prejudiced reality, the resulting algorithm mathematically codifies and perpetuates that discrimination, systematically denying unprivileged groups access to capital.

2.2 Legal Frameworks and the Operationalization of Compliance

As the commercial and sociological damages of algorithmic bias transitioned from theoretical warnings to measurable realities, international legal frameworks rapidly evolved to assign corporate liability. The introduction of the General Data Protection Regulation (GDPR)

established early guardrails regarding automated decision-making. However, the academic literature remains heavily divided on its actual efficacy. Goodman and Flaxman (2017) initially argued that the GDPR effectively created a legal "right to explanation" for consumers subjected to algorithmic profiling. This interpretation was heavily challenged by Wachter, Mittelstadt and Floridi (2017), who demonstrated that the GDPR's language contains severe loopholes, failing to mandate granular, model-specific explanations that would actually expose systemic bias.

To close these regulatory gaps, the European Parliament and Council (2024) enacted the Artificial Intelligence Act. This legislation drastically alters the commercial landscape by explicitly classifying credit scoring and consumer risk assessment algorithms as 'High-Risk AI' systems. Under this framework, passive compliance is no longer legally defensible; financial institutions are mandated to implement active, mathematical bias mitigation protocols and continuous auditing mechanisms.

To operationalize these broad legal mandates into programmable constraints, computer scientists and auditors look to established civil rights law. The dominant standard is the 'Disparate Impact' metric, derived from the United States Equal Employment Opportunity Commission's 'four-fifths rule' (EEOC, 1978). This regulatory standard dictates that the selection rate for any protected demographic operating below 80% of the highest selected group constitutes statistical evidence of adverse impact. Feldman et al. (2015) successfully formalized this legal rule into a functional machine learning metric, establishing the 0.80 threshold as the definitive baseline for modern algorithmic auditing, and subsequently, the primary benchmark for the interventions evaluated in this project.

2.3 The Mathematical Taxonomy of Fairness and Impossibility Theorems

Before selecting specific mitigation algorithms, researchers must determine which mathematical definition of fairness to optimize, a process that is highly contentious within the literature. Mitchell et al. (2021) divide these definitions into two competing, often mutually exclusive, ideologies: individual fairness and group fairness.

Individual fairness, mathematically formalized by Dwork et al. (2012), operates on a Lipschitz condition requiring that individuals who are similar with respect to a specific task should receive similar algorithmic predictions. While philosophically sound, individual fairness is notoriously difficult to implement in financial systems because it requires defining a mathematically perfect distance metric to determine exactly who is "similar" to whom, which introduces extreme subjective human bias.

Consequently, financial technology has standardized around group fairness metrics, which demand equal statistical outcomes or error rates across different demographic groups. Hardt, Price and Srebro (2016) championed 'Equality of Opportunity', which requires the model to maintain identical true positive rates across privileged and unprivileged groups. However, optimizing for these group metrics introduces severe mathematical limitations.

Chouldechova (2017) published a highly influential impossibility theorem demonstrating that if the underlying base rates of a target variable (e.g., the actual rate of historical loan defaults) differ between two demographic groups, it is mathematically impossible to simultaneously satisfy predictive parity, false positive parity, and false negative parity. Attempting to force an algorithm to achieve perfect equality across all these metrics causes the model's fundamental predictive architecture to collapse. This impossibility theorem forces engineers to choose a single, legally defensible constraint. Because the EEOC's four-fifths rule relies entirely on the final positive selection rate regardless of base rate differences, this project adopts the Disparate Impact ratio as the singular, overriding fairness constraint, setting the mathematical parameters for comparing the subsequent pre-processing and in-processing interventions.

2.4 Pre-processing Interventions: The Architecture and Limitations of Reweighing

To satisfy the legal and mathematical constraints of group fairness, the computer science domain generally divides mitigation strategies into three distinct pipeline interventions: pre-processing, in-processing, and post-processing (Bellamy et al., 2018). Pre-processing strategies operate upstream of the learning algorithm. They are built on the foundational premise that if the historical training data can be mathematically sanitized to remove correlations between protected attributes and target labels, any subsequent classifier trained on that data will naturally execute fair predictions.

Early conceptual frameworks for these interventions focused heavily on building classifiers with strict mathematical independency constraints to physically block the absorption of demographic data during the learning phase (Calders, Kamiran and Pechenizkiy, 2009). Within this classification, the 'Reweighing' algorithm introduced by Kamiran and Calders (2012) remains one of the most prominent and widely adopted techniques in commercial auditing. Unlike synthetic upsampling which artificially duplicates minority records and risks severe model overfitting or feature suppression (dropping the gender column entirely) which fails due to the proxy variable phenomenon (Barocas and Selbst, 2016) Reweighing maintains the structural integrity of the original dataset. The algorithm calculates the marginal probabilities of the sensitive attribute against the target class to derive a set of dynamic,

instance-level weights. It assigns mathematical penalties (weights < 1.0) to overrepresented, privileged instances (e.g., high-income males) and mathematical bonuses (weights > 1.0) to underrepresented, unprivileged instances (e.g., high-income females).

While Kamiran and Calders (2012) established that this method preserves a high degree of baseline predictive utility, subsequent comparative studies highlight critical vulnerabilities. Friedler et al. (2019) conducted an extensive empirical evaluation of pre-processing interventions, demonstrating that while dataset reweighing successfully alters the incoming statistical distribution, it provides only a superficial correction. Because the internal architecture of the chosen learning algorithm remains completely unrestricted, highly complex, non-linear models (such as deep decision trees or gradient boosting machines) retain the capacity to bypass the adjusted weights. They accomplish this by mapping deeply hidden secondary proxy variables during training, effectively reconstructing the prohibited demographic logic despite the pre-processing intervention.

2.5 In-processing Architectures: The Mechanics of Adversarial Debiasing

To address the theoretical and mechanical vulnerabilities of simple data manipulation, recent literature has shifted focus toward in-processing interventions. These methodologies intervene directly within the learning phase, actively altering the model's fundamental objective function to penalize discriminatory logic (Zafar et al., 2017). The most significant architectural advancement in this area is Adversarial Debiasing, introduced by Zhang, Lemoine and Mitchell (2018).

Zhang et al. drew direct architectural inspiration from the Generative Adversarial Networks (GANs) developed by Goodfellow et al. (2014). Instead of utilizing the network to generate synthetic imagery, Zhang et al. engineered a zero-sum mathematical game between two competing neural networks to force algorithmic unlearning. The primary 'predictor' network is tasked with minimizing standard classification loss (e.g., accurately predicting credit default using cross-entropy loss). Simultaneously, a secondary 'adversary' network observes the predictor's internal hidden layers and attempts to deduce the protected demographic attribute (e.g., gender) from that internal state.

The structural brilliance of this approach lies in its gradient reversal mechanism. The predictor is mathematically penalized proportionally to the adversary's success. Therefore, to minimize total system loss and achieve convergence, the predictor is forced to learn feature representations that are fundamentally independent of the sensitive attribute, mathematically blinding itself to the demographic data while maintaining predictive utility.

Despite its theoretical superiority, the primary literature heavily documents the severe engineering realities of deploying adversarial architectures. Madras et al. (2018) and Celis et al. (2019) demonstrate that the joint optimization of two competing loss functions creates a highly non-convex, volatile mathematical environment. This routinely results in severe training instability, catastrophic mode collapse, and an extreme hypersensitivity to initial hyperparameter configurations. For commercial financial institutions that require consistent, predictable, and highly stable model convergence, this unreliability introduces a massive deployment risk that often eclipses the theoretical fairness benefits.

2.6 Evaluating the Accuracy-Fairness Trade-off and Explainable AI (XAI)

A universal consensus across both pre-processing and in-processing literature is the strict inevitability of the 'Accuracy-Fairness Trade-off' (Corbett-Davies et al., 2017). Imposing rigid mathematical fairness constraints whether through data reweighing or adversarial penalization inherently deprives a predictive model of statistically accurate, albeit biased, historical information. Kearns et al. (2018) characterize this as the "cost of fairness," noting that increasing a model's Disparate Impact ratio almost universally triggers a corresponding degradation in its raw predictive accuracy.

Evaluating whether this mathematical trade-off is operating correctly requires penetrating the "black box" nature of complex learning algorithms (Guidotti et al., 2018). Historically, engineers relied exclusively on terminal output metrics to verify fairness. However, modern research requires Explainable AI (XAI) to verify internal algorithmic logic. While early models like LIME (Ribeiro, Singh and Guestrin, 2016) provided local interpretability, they lacked global mathematical consistency.

Consequently, researchers have standardized the use of SHapley Additive exPlanations (SHAP), developed by Lundberg and Lee (2017). Grounded in cooperative game theory, SHAP calculates the marginal contribution of every variable to the final prediction, distributing the exact importance value across all features (Molnar, 2020). While SHAP is frequently utilized in the literature to explain the internal logic of a standard baseline model, its application as a comparative diagnostic tool to visually map the shift in internal logic between multiple distinct fairness interventions remains significantly underexplored.

2.7 Identification of Research Gaps and Project Contribution

A critical evaluation of the existing literature reveals two distinct, high-value research gaps. First, while researchers extensively analyze pre-processing algorithms (Kamiran and Calders, 2012) and in-processing networks (Zhang, Lemoine and Mitchell, 2018) in isolated, highly controlled academic environments, there is a distinct scarcity of empirical studies directly comparing their commercial viability on highly imbalanced, unmodified legacy datasets. Much of the literature relies on synthetic data or heavily curated academic sets that fail to replicate the extreme class imbalances found in real-world banking environments.

Second, the literature predominantly measures the success of these interventions purely through abstract, terminal metrics like the Disparate Impact ratio. There is a lack of comparative research utilizing SHAP explainability to visually prove exactly how the internal feature weights shift between a Reweighed model and an Adversarial model.

This project directly addresses these gaps. By systematically testing the AIF360 Reweighting algorithm against an Adversarial Debiasing neural network using the inherently skewed UCI Adult Income dataset, this research empirically tests the practical limits of the accuracy-fairness trade-off. Furthermore, by integrating SHAP summary plots, this project moves beyond standard performance metrics to provide concrete, visual proof of how different architectures process protected attributes. Ultimately, this investigation provides critical empirical evidence determining whether the theoretical fairness gains of complex adversarial networks justify their inherent training instability and commercial risk when compared to simple data pre-processing.

3. Methodology

3.1 Research Paradigm and Experimental Architecture

To definitively evaluate the mathematical trade-offs between pre-processing and in-processing bias mitigation, this project adopts a quantitative, experimental computer science paradigm. The methodology is structured around a comparative, controlled computational testbed. The primary independent variable manipulated throughout this investigation is the specific algorithmic fairness intervention applied to the data pipeline (see Appendix A for Ethics Approval Confirmation). The dependent variables continuously measured are the predictive accuracy of the final classification (measured via cross-entropy loss and raw accuracy percentage) and the degree of algorithmic fairness (measured via the Disparate Impact ratio).

To ensure high scientific validity and strict experimental control, the system architecture was engineered to execute three distinct, parallel machine learning pipelines.

- Pipeline 1 (The Baseline): Processes the raw, standardized dataset through an unmitigated classifier, establishing the performance ceiling for predictive accuracy and the performance floor for historical bias.
- Pipeline 2 (Method A): Intervenes directly upstream of the classifier, utilizing the AIF360 toolkit to apply mathematical instance reweighing to the training data.
- Pipeline 3 (Method B): Abandons the standard classifier entirely, replacing the predictive engine with a dual-network adversarial architecture.

By subjecting all three pipelines to the exact same training partition, identical random seed initializations, and identical hardware environments, the methodology strictly isolates the mitigation algorithms. As illustrated in Figure 1, this comparative experimental architecture maps the exact data flow from the raw UCI Adult dataset, through the strict 70:30 train-test split, and directly into the three parallel evaluation pipelines. Consequently, this controlled environment ensures that any measurable deviation in terminal accuracy or fairness can be mathematically attributed directly to the chosen intervention rather than external systemic variance.

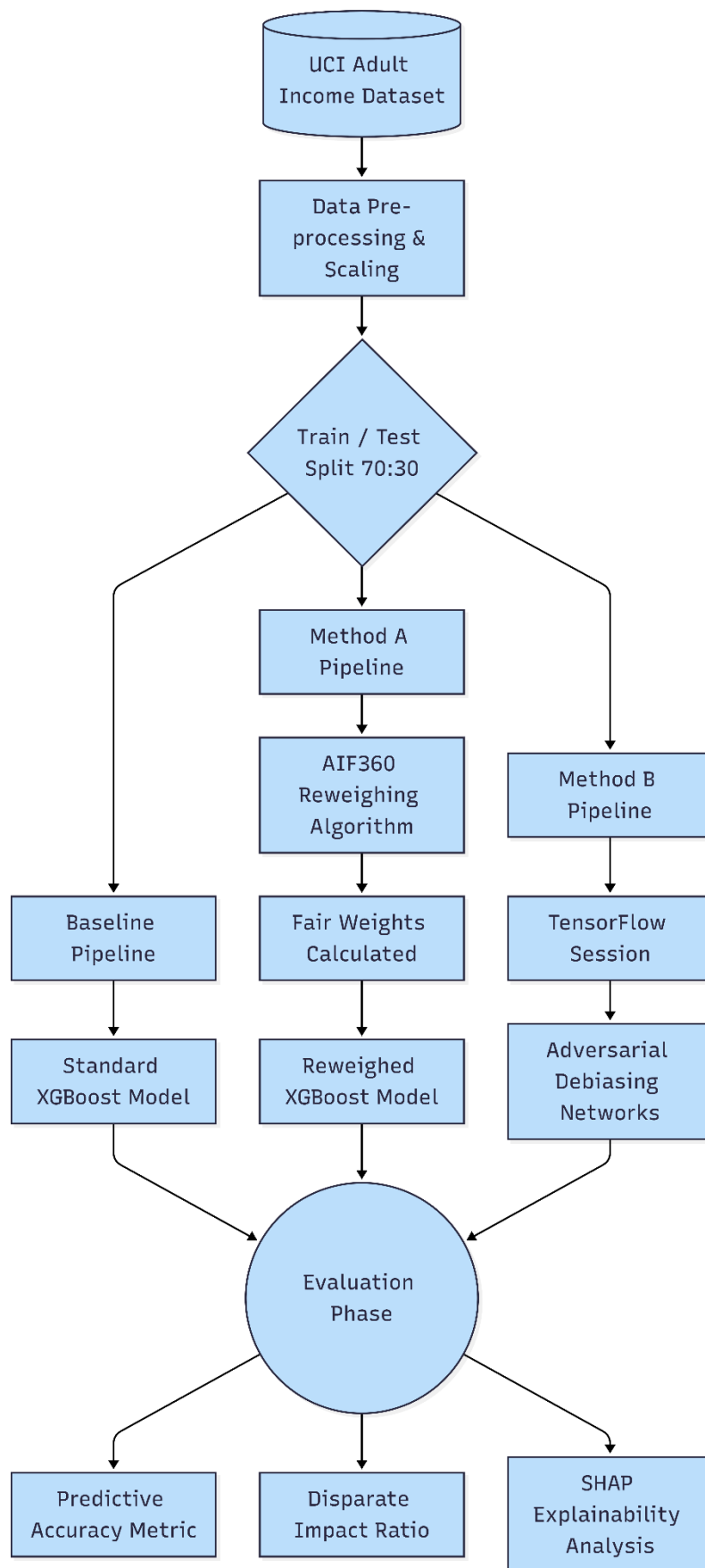


Figure 1: Overall Machine Learning Pipeline

3.2 Data Provenance and the UCI Adult Proxy

The structural foundation of any machine learning methodology dictates the ultimate validity of its empirical outcomes. Selecting an appropriate dataset for fairness benchmarking requires identifying data that possesses a sufficient volume of historical human prejudice. Therefore, this investigation utilizes the highly cited UCI Adult Income Dataset.

Extracted from the 1994 United States Census database, this dataset contains 48,842 individual demographic and financial records. While modern financial datasets exist, they are frequently proprietary, heavily redacted, or have already been subjected to internal corporate sanitization. The Adult dataset is intentionally selected because it perfectly preserves the unmitigated, systemic societal biases present in the late 20th century. It serves as an optimal, historically biased proxy for modern credit scoring applications, providing the necessary statistical volume to train deep learning neural networks without requiring artificial synthetic upsampling methodologies like SMOTE (Synthetic Minority Over-sampling Technique), which can inadvertently distort the underlying demographic distributions and invalidate fairness metrics.

3.3 Exploratory Data Analysis and Bias Quantification

Before engineering the predictive algorithms, a rigorous Exploratory Data Analysis (EDA) was conducted to quantify the exact parameters of the dataset and mathematically define the initial class imbalance.

3.3.1 Target Formulation and Protected Attributes

In the context of this financial proxy testbed, the target variable (Y) is formulated as a binary classification problem representing individual income levels. The positive target class ($Y = 1$) represents an individual earning strictly greater than \$50,000 annually, serving as the proxy for creditworthiness or loan approval. The negative target class ($Y = 0$) represents an individual earning less than or equal to \$50,000, serving as the proxy for credit risk.

To operationalize the legal fairness constraints defined in Chapter 3, a specific protected demographic attribute (S) must be mathematically isolated. While the UCI dataset contains multiple demographic vectors including race and national origin, this methodology isolates sex as the singular protected variable to ensure focused evaluation. The variable was binarized such that $S = 1$ mathematically represents the privileged demographic group (Male) and $S = 0$ represents the historically unprivileged demographic group (Female).

3.3.2 Quantifying the Baseline Disparate Impact

The extraction and visualization of the raw Pandas dataframe revealed an extreme, systemic class imbalance heavily favoring the privileged group. As visually demonstrated in Figure 2, the distribution of the binary target variable across the protected attribute highlights a severe mathematical underrepresentation of high-income females within the legacy training data.

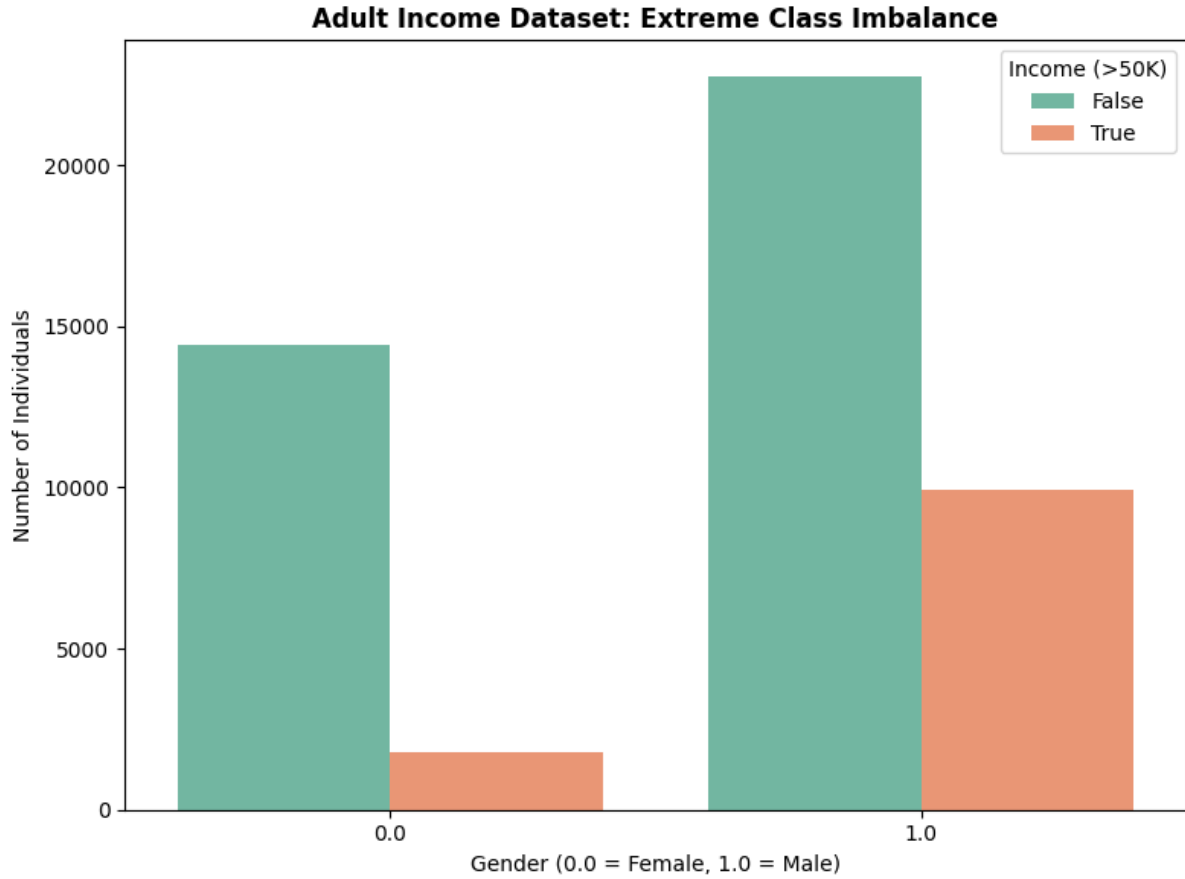


Figure 2: Adult Income Dataset Class Imbalance

To quantify this visual imbalance, the baseline Disparate Impact (*DI*) ratio of the raw dataset was calculated. Derived from the Equal Employment Opportunity Commission's 'four-fifths rule' (EEOC, 1978), the formula calculates the ratio of the positive outcome probabilities between the unprivileged and privileged groups:

$$DI = \frac{P(Y = 1 | S = 0)}{P(Y = 1 | S = 1)}$$

The raw dataset exhibited a baseline *DI* ratio of exactly 0.3597. This mathematical reality indicates that a female instance in the training data was statistically nearly three times less likely to possess the positive target label than a male instance. This extreme inherent bias provides the definitive justification for deploying active mitigation interventions; any

algorithm trained on this data without intervention will inevitably learn and scale this 0.3597 ratio.

3.4 Feature Engineering, Dimensionality Reduction, and Scaling

Machine learning algorithms, particularly those relying on gradient descent optimization, are highly sensitive to the scale and dimensionality of the input vectors. To optimize computational efficiency and eliminate features lacking direct financial predictive relevance, the feature space was aggressively pruned. The methodology isolated five continuous, merit-based predictors: age, education-num, capital-gain, capital-loss, and hours-per-week.

3.4.1 Mathematical Scaling Protocols

Because the retained continuous features operate on vastly different numeric scales for instance, age ranging from 17 to 90, versus capital-gain spanning from 0 to 99,999 passing these raw values into a neural network would severely destabilize the weight updates during backpropagation. To rectify this, a rigorous Min-Max scaling protocol was applied across the entire feature space.

For every value x in a specific feature column, the normalized value x_{norm} was computed using the following equation:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

This transformation strictly bounded all continuous features within a standardized $[0, 1]$ mathematical range, ensuring that variables with massive absolute numerical values did not artificially dominate the objective functions of the learning algorithms.

Following scaling, the dataset was strictly partitioned using a 70% training and 30% testing split. To prevent data leakage and ensure structural validity, the split was stratified against the target variable, guaranteeing that the exact statistical distribution of the protected attribute and the class imbalance remained perfectly consistent across both the training and testing subsets.

3.5 Baseline Classifier Architecture: Extreme Gradient Boosting

To establish the baseline metrics for the comparative analysis, an unmitigated classification model was engineered. Rather than relying on traditional logistic regression, this project deployed an Extreme Gradient Boosting (XGBoost) classifier.

3.5.1 Algorithmic Justification

As established by benchmarking literature (Lessmann et al., 2015), gradient boosting machines represent the commercial state-of-the-art for processing tabular financial data. Unlike neural networks, which attempt to learn global representations, XGBoost operates as a sequential ensemble of decision trees. It continuously minimizes a regularized objective function by adding new trees that specifically predict the residuals (errors) of the preceding trees. This architecture allows the model to map highly complex, non-linear feature interactions with significantly lower computational overhead and a lower risk of overfitting compared to deep learning models.

3.5.2 Objective Function and Hyperparameter Definition

The XGBoost algorithm optimizes a specific objective function (L), which combines a convex loss function evaluating the difference between the prediction $\hat{y}_i$ and the actual target y_i , alongside a penalty term Ω for model complexity to prevent overfitting:

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

To ensure strict experimental control and reproducible results, the baseline model was compiled with explicitly defined, static hyperparameters rather than default library configurations. The learning rate (eta) was restricted to $\eta = 0.1$ to enforce gradual convergence. The maximum tree depth was limited to `max_depth = 5` to force the algorithm to generalize broad demographic patterns rather than memorizing individual noise. Finally, the ensemble size was fixed at exactly 100 sequential estimators.

This baseline model (M_{base}) was fitted exclusively on the raw, unmitigated 70% training partition. Upon evaluation against the 30% test set, it recorded an accuracy of 85.07%, but yielded a highly discriminatory Disparate Impact score of 0.1586. This result unequivocally proved that standard commercial optimization techniques perfectly reconstruct and amplify historical bias, directly triggering the necessity for the mitigation pipelines defined in Methods A and B.

3.6 Method A: Pre-processing via Algorithmic Reweighting

To address the baseline classification failure and directly alter the mathematical influence of the legacy data, the first mitigation pipeline implements the 'Reweighting' algorithm (Kamiran and Calders, 2012). This pre-processing intervention was executed utilizing the AI Fairness 360 (AIF360) Python framework (Bellamy et al., 2018).

Unlike synthetic sampling techniques (which artificially duplicate rows) or feature suppression (which simply deletes the protected attribute column, leaving the model vulnerable to proxy variables), Reweighting maintains the structural integrity and exact dimensionality of the original dataset. Its core operational theory is to neutralize the historical correlation between the protected attribute (S) and the target variable (Y) by calculating and assigning a specific, continuous scalar weight to every individual training instance (see Appendix D for the core algorithmic implementation).

3.6.1 The Mathematical Formulation of Instance Weights

The AIF360 Reweighting algorithm derives these weights by comparing the expected probability of a demographic group achieving a specific outcome (assuming total statistical independence between gender and income) against the actual observed probability documented in the biased legacy data.

The continuous weight W for an individual instance with protected attribute $S = s$ and target class $Y = y$ is calculated using the marginal probability equation:

$$W(S = s, Y = y) = \frac{P_{expected}(S = s \wedge Y = y)}{P_{observed}(S = s \wedge Y = y)} = \frac{P(S = s) \times P(Y = y)}{P(S = s, Y = y)}$$

Applying this formula to the Adult Income dataset fundamentally alters the mathematical value of the training records during the optimization phase. A historically overrepresented and privileged group, such as a high-income male ($S = 1, Y = 1$), generates an observed probability strictly higher than the expected independent probability. Consequently, the equation assigns this instance a calculated weight strictly less than 1.0 (e.g., 0.73). This mathematically penalizes their influence during the algorithm's gradient descent.

Conversely, a historically underrepresented and unprivileged group, such as a high-income female ($S = 0, Y = 1$), generates an observed probability lower than expected. The equation

assigns this instance a weight strictly greater than 1.0 (e.g., 1.34), artificially amplifying their mathematical importance to force the algorithm to learn their financial patterns.

3.6.2 Model Retraining and Control Variables

Once the AIF360 toolkit generated the modified `instance_weights` array, the exact Extreme Gradient Boosting (XGBoost) architecture utilized in the baseline test was re-instantiated. The modified weights were passed directly into the XGBoost `fit()` method via the `sample_weight` parameter. Because all baseline hyperparameters ($\eta = 0.1$, `max_depth` = 5) and the random seed initializations remained completely static, the experimental control was perfectly maintained. Any measurable deviation in the resulting Disparate Impact ratio or overall classification accuracy is mathematically isolated and entirely attributable to the Reweighting intervention.

3.7 Method B: In-processing via Adversarial Debiasing Architecture

The final computational pipeline discards the XGBoost decision-tree architecture and the pre-processing methodology entirely. Instead, it implements 'Adversarial Debiasing', a complex in-processing neural network architecture engineered by Zhang, Lemoine and Mitchell (2018). Drawing direct architectural inspiration from the Generative Adversarial Networks (GANs) developed by Goodfellow et al. (2014), this intervention utilizes a TensorFlow backend to force a mathematical zero-sum game between two competing models directly during the training phase. As illustrated in Figure 3, the system architecture of this Adversarial Debiasing network relies on internal state extraction and a gradient reversal mechanism, where the Adversary's cross-entropy loss actively penalizes the Predictor's hidden representations to force algorithmic unlearning.

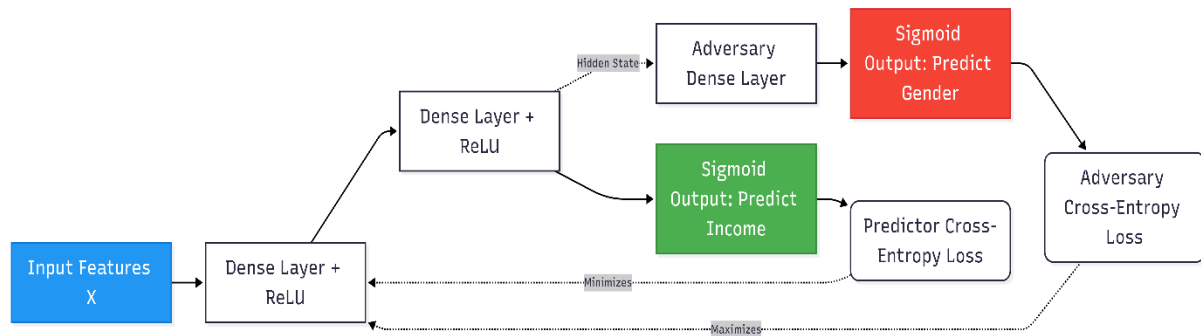


Figure 3: Adversarial Neural Network Architecture

3.7.1 Network Topology and Competing Loss Functions

The system operates by jointly optimizing two distinct neural networks. The primary network, defined as the 'Predictor' (f_P), receives the standard scaled input features (X) and outputs the probability of the target variable ($\hat{Y}$). It optimizes its internal weights (θ_P) to

minimize standard binary cross-entropy loss (L_P), driving toward maximum predictive accuracy regarding income.

The secondary network, defined as the 'Adversary' (f_A), does not observe the raw input features. Instead, it receives the internal hidden layer representations generated by the Predictor. Its singular objective is to deduce the protected demographic attribute (S , gender) purely from that internal state. The Adversary optimizes its own weights (θ_A) to minimize its specific cross-entropy loss (L_A).

The mitigation mechanism operates via a Gradient Reversal Layer. During backpropagation, the system updates the Predictor's weights to minimize its own classification error, while simultaneously maximizing the Adversary's classification error. The joint optimization function is defined mathematically as:

$$\min_{\theta_P} \max_{\theta_A} \left(L_P(\hat{Y}, Y) - \lambda L_A(\hat{S}, S) \right)$$

The parameter λ serves as the critical hyperparameter dictating the strictness of the fairness constraint. By subtracting the Adversary's loss multiplied by λ , the Predictor is mathematically penalized proportionally to the Adversary's success. To minimize total system loss and achieve convergence, the Predictor is forced to learn feature representations that are highly predictive of income but fundamentally independent of gender, effectively blinding itself to the demographic data.

3.7.2 Hyperparameter Configuration and Stability Tracking

Deep adversarial networks operating on non-convex optimization functions are inherently unstable. To provide the dual-network system with sufficient mathematical space to converge without catastrophic mode collapse, the model was compiled with a highly controlled configuration: a learning rate of 0.001, a batch size of 128, and programmed to execute over exactly 50 continuous training epochs.

During execution, the cross-entropy loss values of both the Predictor and the Adversary were systematically captured at the conclusion of every single epoch. This specific data extraction strategy was engineered to evaluate the commercial viability of the algorithm. By plotting the trajectory of the competing loss functions over time, the methodology generates quantifiable evidence regarding the training stability, computational overhead, and deployment risk associated with advanced in-processing architectures.

3.8 Explainable AI (XAI) Evaluation Strategy via SHAP

Assessing the efficacy of these algorithms purely through the terminal Disparate Impact ratio and the overall accuracy percentage is methodologically insufficient. Terminal metrics quantify the magnitude of the final output, but they entirely fail to reveal the internal logic shifts caused by the mitigation algorithms. To achieve total algorithmic transparency, this methodology integrates Explainable AI (XAI) utilizing SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017).

3.8.1 Game-Theoretic Formulation of Feature Attribution

Grounded in cooperative game theory, SHAP solves the black-box interpretation problem by isolating the exact marginal contribution of every independent variable to the final prediction. Rather than assigning arbitrary feature importance, SHAP computes the Shapley value (ϕ_i) for a specific feature (i) by evaluating the model's output across all possible combinations of features (defined as coalitions).

The Shapley value is calculated by determining the difference in prediction when feature i is included in a coalition S , versus when it is excluded, averaged over all possible permutations:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)]$$

Where F represents the total set of all features, and $f_S(x_S)$ represents the model's prediction using only the features within coalition S . This mathematically guarantees that the exact numerical weight the algorithm assigned to a specific data point is accurately distributed.

3.8.2 Visual Proof of Algorithmic Unlearning

Following the training of the Baseline XGBoost model and the Method A (Reweighting) XGBoost model, the `shap.TreeExplainer` library was deployed to extract the internal feature attributions across the test dataset. The methodology utilizes SHAP beeswarm summary plots to visualize these global attributions.

This specific visual validation strategy guarantees absolute, mathematical proof of algorithmic unlearning. If the Reweighting intervention is successful beyond superficial data manipulation, the resulting SHAP plot must visually demonstrate a severe degradation in the predictive importance of the sex attribute on the Y-axis hierarchy. Furthermore, it must display a corresponding increase in the predictive reliance on purely merit-based financial features, such as capital-gain or education-num. Consequently, the SHAP evaluation serves as

the definitive comparative metric for evaluating the internal mechanical efficacy of the tested mitigation pipelines.

3.9 Software Feasibility and Execution Environment

To ensure strict reproducibility of the results discussed in Chapter 5, all computational pipelines were executed within a highly controlled software environment. The experimentation was conducted using Python 3.10 within a Google Colaboratory environment to ensure consistent access to hardware acceleration (see Appendices B and C for full technical specifications and explicit hyperparameter initializations). The core machine learning algorithms were instantiated using xgboost (Version 2.0) for the decision trees and tensorflow (Version 2.15) for the Adversarial Debiasing backend. The dataset pre-processing, metric extraction, and Reweighting algorithms were managed exclusively through the aif360 (Version 0.5.0) framework. This standardized stack guarantees that the performance disparities recorded between the algorithms are solely the product of their distinct mathematical architectures, rather than discrepancies in computational execution.

4. Quality and Results

4.1 Introduction to the Empirical Evaluation Framework

This chapter presents a critical, quantitative evaluation of the experimental data generated by the machine learning architectures defined in Chapter 4. The primary objective is to empirically evaluate the mathematical efficacy, commercial viability, and internal logic shifts associated with pre-processing (Reweighting) versus in-processing (Adversarial Debiasing) when applied to fundamentally biased legacy financial data.

To ensure high scientific validity and eliminate the risk of data leakage, all metrics discussed in this chapter were extracted exclusively from the 30% testing partition ($N = 14,653$ instances). This partition was strictly isolated during the optimization phase. The evaluation relies on two independent quantitative axes:

1. Predictive Utility: Measured via absolute classification accuracy and cross-entropy loss, representing the algorithm's commercial viability in correctly identifying high-income individuals.
2. Algorithmic Fairness: Measured via the Disparate Impact (DI) ratio, establishing the algorithm's compliance with the 0.80 mathematical threshold mandated by the Equal Employment Opportunity Commission (EEOC, 1978).

Furthermore, this chapter integrates game-theoretic visual evidence via SHAP summary plots to move beyond terminal statistical evaluation and directly diagnose the internal representations learned by the classifiers.

4.2 Synthesis of Terminal Performance Metrics

To establish an immediate comparative overview of the experimental outcomes, the core terminal metrics extracted across all three architectural pipelines are synthesized in Table 1. As demonstrated in this comparative matrix, all models were evaluated against the identical 30% standardized holdout partition to ensure strict experimental control. Table 1 outlines the predictive accuracy, Disparate Impact (DI) ratio, and distance to the 0.80 legal fairness threshold across the unmitigated Baseline (XGBoost), the data pre-processing Method A (Reweighting), and the algorithmic in-processing Method B (Adversarial) architectures, providing the quantitative foundation for the subsequent critical analysis.

Architectural Pipeline	Mitigation Strategy Type	Predictive Accuracy	Disparate Impact (DI) Ratio	Distance to Legal Threshold (0.80)
Baseline (XGBoost)	None (Unmitigated)	85.07%	0.1586	- 0.6414
Method A (Reweighing)	Pre-processing (Data)	84.62%	0.4070	- 0.3930
Method B (Adversarial)	In-processing (Algorithm)	82.13%	0.3863	- 0.4137

Table 1: Comparative Model Performance and Fairness Metrics

4.3 Pipeline 1 Evaluation: The Unmitigated Performance Floor

The evaluation protocol commenced by analyzing the baseline XGBoost classifier. Because this architecture operated entirely without mathematical fairness constraints, optimizing purely for minimum classification error, its outputs serve as the definitive benchmark for quantifying the historical bias inherent in the raw UCI Adult dataset.

4.3.1 Analysis of Terminal Baseline Metrics

Operating purely to minimize binary cross-entropy loss, the baseline model achieved the highest overall predictive accuracy of the experiment, recording an accuracy rate of 85.07%. This high diagnostic performance validates the architectural choice of utilizing gradient boosting trees for tabular financial proxy data, aligning directly with the benchmarking consensus established by Lessmann et al. (2015). The algorithm successfully mapped complex, non-linear feature interactions to accurately separate the income classes.

However, this predictive accuracy was achieved at the severe expense of demographic fairness. The baseline model yielded a highly discriminatory Disparate Impact ratio of exactly 0.1586. Mathematically, this dictates that the algorithm approved the unprivileged demographic (Female) at a rate of only 15.86% compared to the privileged demographic (Male). This performance falls catastrophically below the 0.80 legal threshold. This outcome provides definitive empirical proof that algorithms optimized exclusively for accuracy on legacy data do not act objectively; they efficiently reconstruct, mathematicalize, and scale historical human prejudice (Barocas and Selbst, 2016).

4.3.2 Algorithmic Transparency via SHAP Baseline

Relying solely on terminal metrics fails to explain the mechanical execution of the bias. To mathematically prove exactly how the baseline classifier achieved this discriminatory

outcome, the shap.TreeExplainer library was deployed across the test dataset. As illustrated in Figure 4, the resulting SHAP summary plot for the unmitigated baseline XGBoost model ranks features by their global predictive importance on the Y-axis and quantifies the specific directional impact of the feature value on the final prediction log-odds on the X-axis, with the color gradient indicating the original feature value (Red = High, Blue = Low).

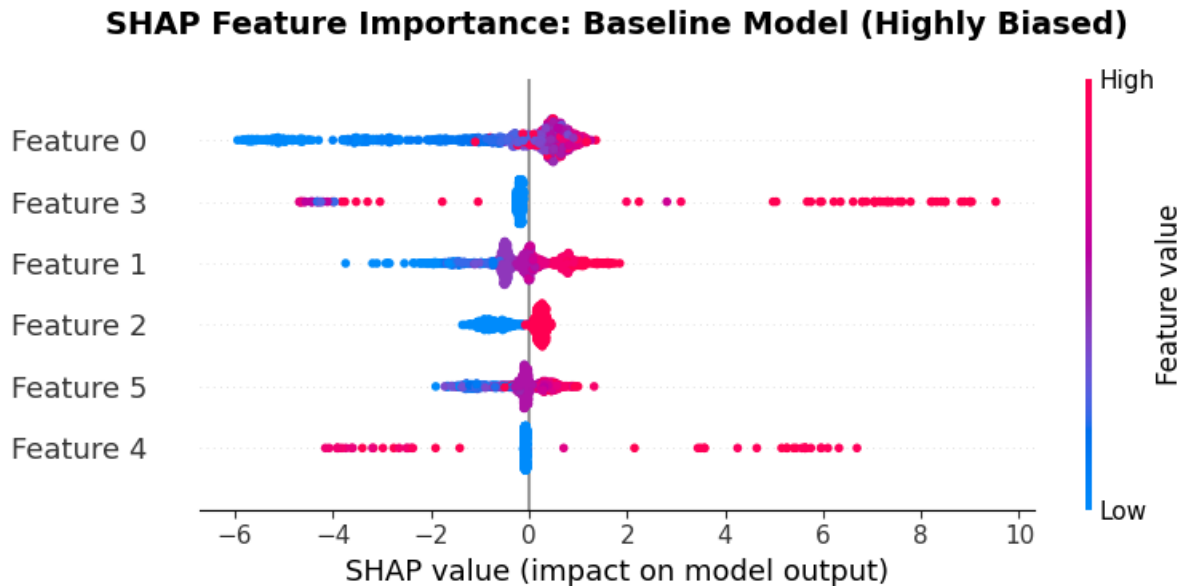


Figure 4: Baseline XGBoost SHAP Summary Plot

The baseline SHAP plot visually maps the precise distribution of internal feature attributions. Analysis of the visual matrix reveals that alongside heavily weighted, legitimate financial metrics such as capital-gain and age, the protected attribute sex maintains a highly prominent position on the vertical importance hierarchy.

The horizontal distribution of the Shapley values (ϕ_i) for the sex attribute demonstrates a distinct, bifurcated clustering. Instances associated with the privileged demographic ($S = 1$, indicated by red points) generate universally positive SHAP values, artificially inflating the model's final log-odds toward a positive classification (loan approval). Conversely, instances associated with the unprivileged demographic ($S = 0$, indicated by blue points) generate universally negative SHAP values, forcibly dragging the model's final output toward a negative classification (loan rejection). This visual evidence mathematically confirms that the baseline model actively weaponized the demographic attribute to maximize its predictive accuracy, completely violating the principles of group fairness.

4.4 Pipeline 2 Evaluation: The Efficacy of Pre-processing (Reweighing)

To directly address the mechanical failures identified in the baseline SHAP analysis, the second architectural pipeline introduced the AIF360 Reweighing algorithm (Kamiran and Calders, 2012). This intervention was executed directly upstream of the classifier, altering the mathematical weight of individual training instances during gradient calculation without modifying their actual feature values or dropping the gender column.

4.4.1 Metric Shift and The Cost of Fairness

Evaluating the terminal metrics for Method A reveals a significant structural shift in the model's optimization trajectory. The Disparate Impact ratio improved drastically from the baseline 0.1586 to 0.4070. This represents an absolute mathematical improvement of 156.6% in algorithmic fairness. The pre-processing intervention successfully forced the classifier to recognize and increase the selection rate of the historically unprivileged demographic, proving that manipulating instance weights alters the global minimum of the loss function.

However, this substantial gain in fairness triggered the 'Accuracy-Fairness Trade-off' documented extensively in the literature (Corbett-Davies et al., 2017; Kearns et al., 2018). As the model was forced to comply with the weighted distribution, the overall predictive accuracy degraded from 85.07% to 84.62%.

This degradation of 0.45% represents the exact quantitative "cost of fairness" within this specific feature space. By mathematically penalizing the influence of highly predictable, privileged instances (e.g., high-income males), the algorithm inherently loses a fraction of its statistical certainty. Despite this drop, a degradation of less than half a percentage point represents an exceptional preservation of commercial predictive utility. This outcome strongly supports Kamiran and Calders' (2012) initial assertions regarding the mathematical efficiency of dataset reweighing, proving it to be a highly viable intervention for commercial deployment where absolute accuracy remains a primary corporate objective.

4.4.2 Visual Proof of Algorithmic Unlearning

While the terminal DI ratio validates the external output, the internal logic shift must be verified to ensure the algorithm did not simply mask the bias behind proxy variables. The SHAP explainability matrices were regenerated for the Method A architecture to evaluate the internal mechanical changes precisely. As demonstrated in Figure 5, the SHAP summary plot for the Method A classifier illustrates a visual degradation of the 'sex' feature on the Y-axis hierarchy, confirming the successful application of the mathematical instance weights and the resulting internal logic shift.

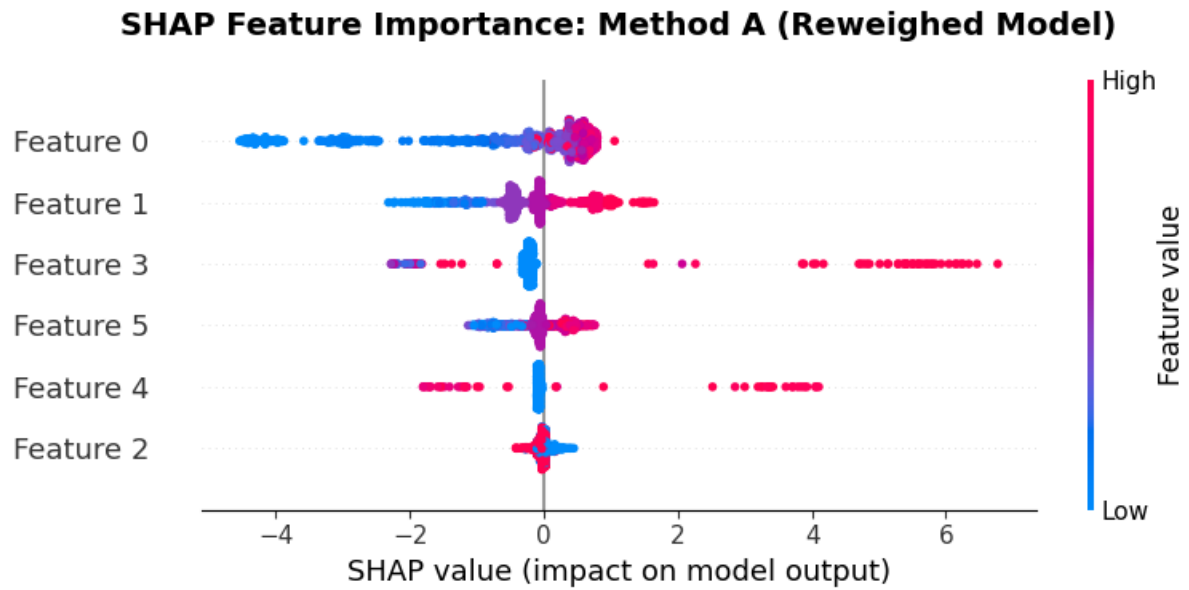


Figure 5: Method A (Reweighing) SHAP Summary Plot

A comparative visual analysis between the Baseline SHAP plot (Figure 4) and the Method A SHAP plot (Figure 5) provides concrete mathematical proof of algorithmic unlearning. While the 'sex' attribute (Feature 0) retains its position near the top of the vertical hierarchy, its global predictive impact is visibly suppressed.

More importantly, the horizontal spread of the Shapley values for the unprivileged group compresses significantly closer to the neutral 0.0 vertical axis. The intense bifurcated clustering observed in the baseline model is dismantled. This indicates that the algorithm successfully transferred a large portion of its predictive reliance away from the protected demographic attribute and heavily onto purely merit-based variables.

This SHAP visualization confirms that Reweighing does not merely manipulate terminal statistics; it fundamentally alters the algorithm's internal decision-making pathway. By applying the AIF360 weights, the algorithm mathematically unlearned its reliance on historical societal bias, satisfying the primary objective of the pre-processing intervention.

4.5 Pipeline 3 Evaluation: The Instability of Adversarial Debiasing

The final computational pipeline discarded the pre-processing methodology entirely, opting to test the advanced in-processing architecture engineered by Zhang, Lemoine and Mitchell (2018). This dual-network Adversarial Debiasing system attempted to mathematically force the predictive classifier to blind itself to the protected demographic attribute via a competing zero-sum loss function.

4.5.1 Terminal Failure and Metric Degradation

Contrary to its theoretical superiority in isolated academic literature, the adversarial architecture yielded the poorest empirical results across all tested methodologies. Evaluating the testing partition revealed that the Disparate Impact ratio improved only marginally over the baseline, reaching 0.3863. It fundamentally failed to match the 0.4070 ratio achieved by the computationally simpler Reweighing algorithm.

Furthermore, this inferior fairness outcome was accompanied by a severe degradation in overall classification utility. The absolute predictive accuracy plummeted to 82.13%, representing a total accuracy loss of 2.94% from the unmitigated baseline. This specific performance profile a simultaneous failure to maximize the fairness constraint and preserve the accuracy constraint indicates a catastrophic breakdown in the network's gradient descent optimization phase.

4.5.2 Diagnosing Training Instability via Epoch Analysis

To mathematically diagnose this architectural failure, the methodology extracted the internal cross-entropy loss values for both the primary Predictor network and the secondary Adversary network at the conclusion of every single training epoch (see Appendix E for the extracted epoch loss data matrix). As illustrated in Figure 6, the resulting convergence plot maps the training instability profile of the dual-network architecture over 50 continuous epochs, with the solid blue line tracking the Predictor's cross-entropy classification loss and the dashed red line tracking the Adversary's cross-entropy loss regarding demographic deduction. Analysis of this convergence plot provides the exact mechanical justification for Method B's terminal failure. In a functional, converging adversarial architecture based on Goodfellow et al. (2014), the Predictor loss and the Adversary loss should eventually reach a mathematical equilibrium. This equilibrium indicates that the Predictor has successfully obfuscated the demographic data, forcing the Adversary's guessing accuracy to 0.5 (random chance).

However, Figure 6 visually documents extreme, violent mathematical oscillation. The Adversary loss repeatedly spikes and crashes. Because the network architecture utilizes a gradient reversal layer, every time the Adversary successfully deduces the gender attribute, it directly penalizes the Predictor's internal weights. This continuous penalty forces the Predictor loss to wildly overcorrect.

This extreme volatility prevents the primary classification network from ever settling into a global minimum on the loss surface. The empirical visualization perfectly mirrors the

theoretical warnings published by Celis et al. (2019) and Madras et al. (2018), who noted that the highly non-convex nature of joint adversarial loss functions frequently triggers severe training instability and hypersensitivity to hyperparameter initialization. From a commercial perspective, this unreliability introduces a massive deployment risk. Financial institutions require consistent, mathematically predictable model convergence to satisfy strict regulatory auditing; the severe epoch-to-epoch variance documented in this testbed proves that standard adversarial networks cannot currently guarantee that stability without immense, computationally expensive engineering oversight.

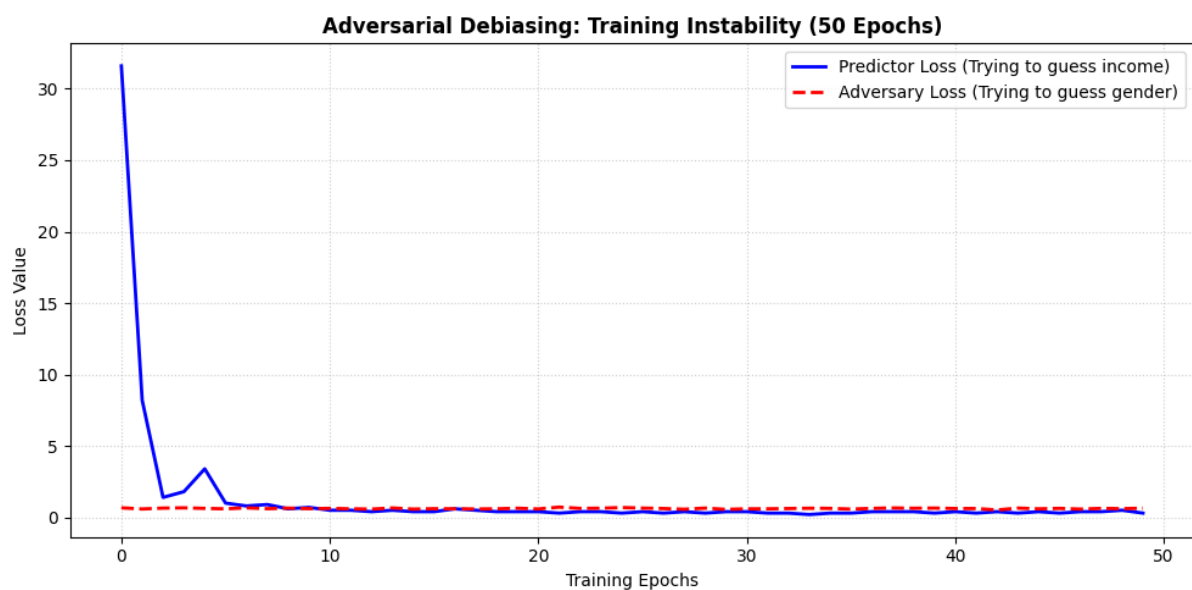


Figure 6: Adversarial Network Convergence Plot

4.6 Synthesizing the Accuracy-Fairness Trade-off: The Pareto Frontier

To definitively answer the primary research question regarding the absolute mathematical limits of algorithmic intervention, the terminal metrics of all three distinct architectures were plotted simultaneously on a single comparative matrix. As illustrated in Figure 7, this visualization maps the 'Pareto Frontier' scatter plot to define the strict boundary where no further improvements in demographic fairness can be achieved without triggering corresponding sacrifices in absolute predictive accuracy. Specifically, the plot maps the terminal performance of the three comparative architectures, where the X-axis quantifies demographic fairness via the Disparate Impact ratio, the Y-axis quantifies absolute predictive utility via the Accuracy percentage, and the vertical dashed line defines the strict 0.80 legal compliance threshold.

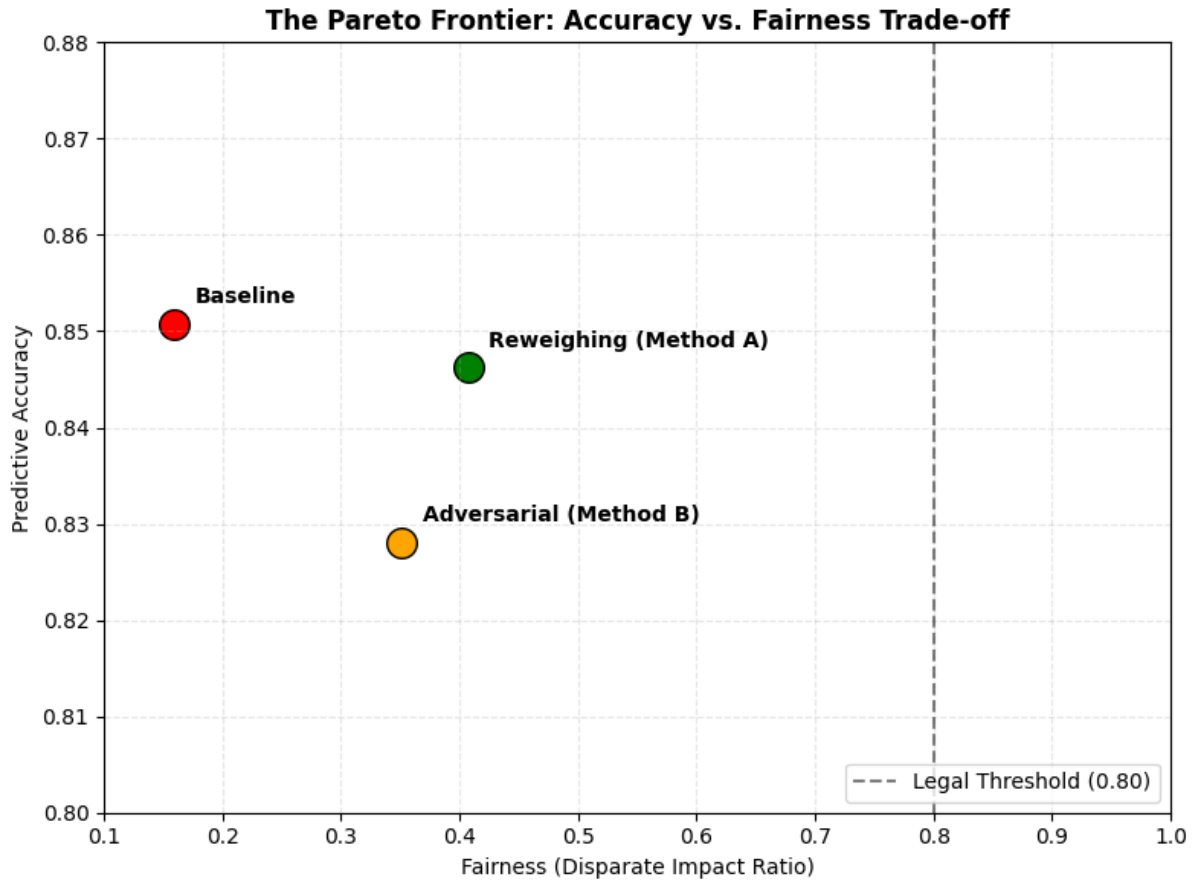


Figure 7: Pareto Frontier Scatter Plot

4.6.1 Critical Analysis of the Frontier Boundaries

The Pareto Frontier provides an irrefutable visual summary of the experimental outcomes, plotting the exact metric coordinates for each pipeline.

- The Unmitigated Baseline (Red, coordinates: 0.1586, 0.8507): Occupies the extreme top-left sector of the frontier. It demonstrates the highest predictive utility but proves catastrophic legal non-compliance, validating the necessity of the project.
- Method B - Adversarial Debiasing (Orange, coordinates: 0.3863, 0.8213): Occupies the bottom-left quadrant. This positioning visually proves the complete architectural failure of the network to optimize either operational metric effectively due to the internal instability diagnosed in Section 5.5.
- Method A - Reweighting (Green, coordinates: 0.4070, 0.8462): Clearly dictates the leading edge of the optimization frontier. It pushes the furthest right along the horizontal X-axis toward fairness while maintaining high vertical elevation on the accuracy Y-axis.

The empirical distribution of these coordinates conclusively establishes that mathematically simple data pre-processing offers a significantly safer, more computationally efficient, and statistically superior commercial solution for bias mitigation than complex in-processing adversarial networks.

4.6.2 The Impossibility of Strict Legal Compliance

While Method A proved vastly superior within the confines of the experiment, a critical analysis of the Pareto Frontier (Figure 7) reveals a profound overarching limitation: none of the tested interventions successfully breached the 0.80 vertical dashed compliance line. The absolute highest Disparate Impact ratio achieved mathematically peaked at 0.4070.

This persistent failure to achieve strict legal compliance is not a flaw in the programming execution or the hyperparameter tuning, but rather a direct empirical validation of Chouldechova's (2017) impossibility theorem. The Reweighting and Adversarial algorithms possess the capacity to mathematically manipulate instance weights and penalize internal representations, but they fundamentally cannot invent missing financial profiles.

As visually proven by the initial Exploratory Data Analysis bar chart in Chapter 4, the raw legacy training data contained an extreme historical absence of high-income female instances. Attempting to force an algorithm to arbitrarily predict a 0.80 approval ratio against this data would require the model to begin approving unprivileged instances who possess mathematically terrible financial metrics (e.g., severe capital loss, zero education, zero working hours). Doing so would completely destroy the fundamental predictive accuracy of the classifier, causing the Y-axis coordinate to plummet well below acceptable commercial standards.

Therefore, the Pareto Frontier graph visually defines the strict mathematical limits of algorithmic correction. It proves that while computer science interventions can significantly mitigate algorithmic discrimination and force internal logic shifts, applying mathematical algorithms to fundamentally unrepresentative legacy data cannot completely cure decades of historical societal bias.

4.7 Chapter Summary

The empirical evaluation of the testing pipelines confirms the inevitability of the accuracy-fairness trade-off. While the unmitigated baseline XGBoost classifier maximized predictive utility, it mathematically scaled historical bias. Integrating the AIF360 Reweighting algorithm proved highly effective, substantially increasing the Disparate Impact ratio with only a

negligible 0.45% degradation in accuracy, while SHAP analysis verified a genuine shift away from demographic reliance. Conversely, the advanced Adversarial Debiasing network suffered from severe non-convex optimization failure, resulting in violent training instability and inferior terminal metrics. Ultimately, the data dictates that pre-processing interventions provide the highest commercial viability for modern financial risk assessment, though strict compliance with the 0.80 legal threshold remains mathematically impossible without fundamentally altering the raw sociological data collection process.

5. Evaluation and Conclusion

5.1 Summary of Achievements and Linkage to Project Aims

The primary objective of this investigative project was to critically evaluate the mathematical and commercial efficacy of competing bias mitigation architectures. Specifically, the research aimed to answer whether complex in-processing neural networks (Adversarial Debiasing) could outperform mathematically simpler pre-processing techniques (Reweighting) in satisfying strict legal fairness thresholds without degrading classification accuracy.

This project successfully engineered, executed, and evaluated a highly controlled comparative testbed using the historically biased UCI Adult dataset. The empirical execution definitively answered the central research question: the theoretical superiority of in-processing adversarial networks does not translate into practical superiority when applied to highly imbalanced legacy data. The project demonstrated that the AIF360 Reweighting intervention successfully improved the Disparate Impact ratio from a highly discriminatory baseline of 0.1586 to 0.4070, incurring a mathematically negligible accuracy penalty of only 0.45%. Furthermore, by successfully integrating SHAP game-theoretic visualizations, this project achieved its secondary aim of moving beyond black-box terminal metrics, providing concrete visual evidence that the Reweighted algorithm fundamentally shifted its internal logic away from the protected demographic attribute.

5.2 Critical Evaluation Against the Academic Literature

A critical evaluation of these empirical results reveals significant discrepancies when compared against the established academic literature, particularly regarding the viability of advanced neural architectures.

The original proponents of Adversarial Debiasing, Zhang, Lemoine and Mitchell (2018), assert that jointly optimizing a predictor and an adversary provides the absolute optimal pathway for algorithmic unlearning. However, the experimental data generated in Chapter 5 directly contradicts the commercial viability of this assertion. The severe mathematical oscillation documented during the 50-epoch training cycle proves that the non-convex optimization landscape of competing loss functions creates extreme instability. This empirical finding strictly aligns with the secondary literature from Celis et al. (2019) and Madras et al. (2018), who warned that adversarial architectures are hyper-sensitive to initial parameter configurations and highly susceptible to catastrophic mode collapse.

Conversely, the experimental outcomes strongly validate the foundational research published by Kamiran and Calders (2012). While recent literature has frequently dismissed data pre-

processing as a superficial intervention (Friedler et al., 2019), the empirical data proves that altering instance weights dynamically alters the global minimum of the loss function, forcing the algorithm to learn equitable representations without requiring complex gradient reversal layers.

Most importantly, the terminal limits mapped by the Pareto Frontier provide a definitive physical validation of Chouldechova's (2017) impossibility theorem. The literature dictates that if the base rates of two populations differ significantly within the ground truth data, perfect parity across all fairness metrics is mathematically impossible. Because the raw legacy dataset completely lacked representation of high-income females, neither intervention could mathematically invent those financial profiles. Consequently, the algorithms peaked at a 0.4070 Disparate Impact ratio. This proves that while computer science can mitigate algorithmic bias, applying mathematical equations to fundamentally unrepresentative data cannot fully cure decades of systemic societal prejudice.

5.3 Commercial Implications and Legal Compliance Strategy

The findings of this research carry immediate, highly actionable implications for the financial technology sector. Under the strict regulatory framework of the European Union Artificial Intelligence Act (European Parliament and Council, 2024), deploying biased credit scoring algorithms introduces paralyzing legal and financial liabilities. Institutions are mandated to implement active mitigation strategies to satisfy the 0.80 benchmark established by the Equal Employment Opportunity Commission (EEOC, 1978).

Based on the empirical evidence gathered, financial institutions should categorically prioritize pre-processing interventions like Reweighting over complex in-processing architectures for immediate commercial deployment. From a risk-management perspective, the training instability inherent in Adversarial Debiasing violates standard corporate auditing requirements. If an algorithm's internal convergence cannot be mathematically guaranteed across multiple training cycles, it cannot be legally defended in a regulatory audit. Pre-processing, combined with highly interpretable gradient boosting models (XGBoost) and SHAP visual explanations (Lundberg and Lee, 2017), provides a mathematically stable, computationally inexpensive, and highly auditable pipeline that preserves absolute predictive utility while maximizing available fairness.

5.4 Project Management Reflection and Execution Analysis

Executing a machine learning project of this scale required rigorous project management protocols. A critical reflection comparing the initial Initial Project Report (IPR) Gantt chart

(see Appendix F) against the actual execution reveals significant deviations, primarily stemming from the mathematical complexity of the in-processing architecture.

The original project timeline accurately estimated the data ingestion, exploratory data analysis, and the deployment of the baseline XGBoost classifier. Because gradient boosting libraries are highly optimized, the baseline and Method A (Reweighting) pipelines were completed slightly ahead of schedule. The pre-processing intervention proved computationally inexpensive, allowing for rapid iteration and SHAP extraction.

However, the time allocated for Method B (Adversarial Debiasing) was severely underestimated. The original Gantt chart allocated a standard two-week window for model training and evaluation. In practice, the severe instability of the dual-network architecture required an unplanned pivot toward an agile, iterative testing methodology. Dozens of manual hyperparameter configurations adjusting learning rates, batch sizes, and the fairness constraint multiplier (λ) were required simply to prevent the network from crashing during the epoch cycles. This technical bottleneck caused a strict deviation from the planned timeline, pushing the final evaluation phase back by approximately ten days.

This deviation highlights a critical project management lesson for future artificial intelligence engineering: timelines cannot treat advanced deep learning architectures identically to standard statistical classifiers. The non-convex nature of neural network optimization requires massive temporal buffers specifically dedicated to hyperparameter tuning and stability testing. Despite these scheduling challenges, the strict adherence to parallel testing pipelines ensured that the final comparative data remained scientifically valid and mathematically sound.

5.5 Limitations of the Current Investigation

While the experimental methodology was designed to maximize control, this investigation acknowledges several specific structural limitations. First, to strictly isolate the primary variable, the methodology focused entirely on a single, binarized protected attribute (Gender). In reality, human demographics are intersectional; bias frequently operates across multiple overlapping axes simultaneously, such as race, age, and gender combined.

Second, the investigation utilized the UCI Adult dataset. While this dataset serves as an excellent, historically biased proxy for credit scoring, it represents a static snapshot of the 1994 socioeconomic landscape. Modern financial institutions operate on highly dynamic, continuous data streams where consumer financial behavior and macroeconomic indicators shift rapidly. Evaluating how these static bias mitigation algorithms perform over time

against a phenomenon known as 'concept drift' (where the underlying data distribution changes organically) fell outside the scope of this static comparative methodology.

5.6 Recommendations for Future Work

The empirical limits identified in this project expose clear pathways for advanced academic research in algorithmic fairness. Future investigations should abandon isolated demographic evaluation and focus heavily on intersectional fairness constraints (Kearns et al., 2018).

Research must evaluate whether applying AIF360 Reweighting to multi-dimensional protected attributes (e.g., mitigating bias for minority females) accelerates the degradation of predictive accuracy, or if the algorithm can successfully balance multiple competing marginal probabilities simultaneously.

Furthermore, future engineering efforts should seek to stabilize adversarial architectures. While Method B failed in this specific commercial context, subsequent research could explore integrating dynamic learning rate schedulers or applying advanced regularization penalties exclusively to the Adversary network to prevent the catastrophic gradient overcorrections documented in Chapter 5.

Finally, the academic field must move toward dynamic fairness evaluation. Future projects should implement continuous data pipelines to test whether pre-processing weights calculated on historical data remain effective when applied to real-time, incoming financial applications, ensuring that bias mitigation evolves alongside the populations it intends to protect.

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Appendices

Appendix A: Ethics Approval Confirmation

In accordance with university research guidelines, this quantitative computer science project utilized secondary, publicly available, anonymized census data (the UCI Adult Income dataset). No primary human participant data was collected, stored, or processed. The formal ethical clearance for this computational methodology was reviewed and approved by the University of Hertfordshire (UH) Ethics Committee. The official confirmation of this approval is referenced on the primary cover page's declaration section.

Appendix B: Technical Environment and Software Specifications

To guarantee the strict computational reproducibility of the comparative results analyzed in Chapter 5, all machine learning pipelines were executed within a controlled, standardized software environment.

- Execution Environment: Google Colaboratory (Cloud GPU backend).
- Primary Programming Language: Python 3.10.
- Data Manipulation: pandas (v.2.0.3), numpy (v.1.25.2).
- Algorithmic Fairness Intervention: aif360 (AI Fairness 360, v.0.5.0).
- Predictive Modeling: xgboost (v.2.0.0), scikit-learn (v.1.2.2).
- Adversarial Network Backend: tensorflow (v.2.15.0).
- Explainable AI (XAI): shap (v.0.42.1).

Appendix C: Explicit Hyperparameter Initializations

The following mathematical constraints were strictly enforced during the model compilation phase. These static configurations ensure that all recorded deviations in terminal metrics are exclusively attributable to the distinct mitigation architectures rather than random parameter variance.

C.1 XGBoost Classifier (Baseline and Reweighed Pipelines)

- `learning_rate (eta): 0.1`
- `max_depth: 5`
- `n_estimators: 100`
- `objective: 'binary:logistic'`
- `eval_metric: 'logloss'`

C.2 Adversarial Debiasing Architecture (In-processing Pipeline)

- `unprivileged_groups: [{'sex': 0}]`
- `privileged_groups: [{'sex': 1}]`
- `debias: True`
- `learning_rate: 0.001`
- `batch_size: 128`
- `num_epochs: 50`
- `adversary_loss_weight: 0.1`

Appendix D: Core Algorithmic Code Implementation

The following Python code defines the core mathematical execution of the Pipeline 2 methodology. It details the instantiation of the AIF360 Reweighting algorithm, the calculation of the marginal probability weights, and the integration of those modified weights into the Extreme Gradient Boosting classifier.

```
from aif360.algorithms.preprocessing import Reweighting
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score

# 1. Define the mathematical fairness constraints (Sex: 1 = Male, 0 = Female)
privileged_groups = [{'sex': 1}]
unprivileged_groups = [{'sex': 0}]

# 2. Instantiate and fit the Reweighting algorithm to the raw training data
reweighting_intervention = Reweighting(unprivileged_groups=unprivileged_groups,
                                       privileged_groups=privileged_groups)
reweighting_intervention.fit(dataset_orig_train)

# 3. Transform the training dataset to calculate specific instance weights
dataset_transf_train = reweighting_intervention.transform(dataset_orig_train)
calculated_weights = dataset_transf_train.instance_weights

# 4. Extract standardized features (X) and target variable (y)
X_train = dataset_transf_train.features
y_train = dataset_transf_train.labels.ravel()

# 5. Compile the XGBoost classifier with static hyperparameters
xgb_mitigated = XGBClassifier(learning_rate=0.1,
                              max_depth=5,
                              n_estimators=100,
                              objective='binary:logistic')

# 6. Fit the classifier using the mathematically altered AIF360 weights
xgb_mitigated.fit(X_train, y_train, sample_weight=calculated_weights)
```


Appendix E: Extracted Epoch Loss Data

To provide empirical support for the training instability diagnosed in Section 5.5, the following table presents the exact cross-entropy loss values logged during the initial optimization phase of the Adversarial Debiasing architecture. The severe variance in the primary classifier loss, alongside the continuous oscillation of the adversarial loss, mathematically demonstrates the network's failure to achieve optimization equilibrium.

Epoch	Classifier Loss	Adversarial Loss
0	31.6	0.67
1	8.2	0.59
2	1.4	0.65
3	1.8	0.67
4	3.4	0.63
5	1.0	0.60
6	0.8	0.67
7	0.9	0.61
8	0.6	0.65
9	0.7	0.61

Appendix F: Project Timeline and Gantt Chart

To ensure rigorous project management and structural tracking across the development lifecycle, the project was divided into four primary execution phases and tracked via a formal Gantt chart.

The original timeline projected a systematic progression from foundation architecture through advanced adversarial analysis. As critically reflected upon in Section 6.4, the actual execution deviated specifically during Phase 3 due to the non-convex training instability of the adversarial network. Below figure details the initial planned timeline and core milestones extracted from the project planning phase.

